# NIKHIL RAJ DEEP


## CONTACT

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## EDUCATION

**PhD in Soil, Water, and Ecosystem Sciences (Pursuing)** — Approx. Graduation Date – Dec, 2029
University of Florida, USA — **GPA – 4/4**
**Thesis Topic/Title:** *Remote sensing of soil moisture, hydraulic soil properties and water fluxes at the field scale.*
**Advisor**: Dr. Ebrahim Babaeian

**Master of Technology in Remote Sensing** — June 2018
Birla Institute of Technology, Mesra, India — CGPA - 8.77/10
**Thesis Topic/Title:** *Impact of climate change on surface water bodies of Ganga River basin - a remote Sensing approach.*
**Advisor**: Dr. Virendra Singh Rathore

**Bachelor of Technology in Civil Engineering** — August 2015
Dhaneswar Rath Institute of Engineering and Management Studies, Cuttack, India — CGPA - 7.14/10
**Thesis Topic/Title:** *Morphometric Analysis of Subarnarekha River using Remote Sensing and GIS technique.*
**Advisor**: Swapna Acharya

## RESEARCH EXPERIENCE

Graduate Research Assistant — Jan 2025 – Till Date
**University of Florida,** USA

- Automated a radiative transfer model for soil moisture estimation at different soil depths for Florida.
- Developed a semi-automated model in Google Earth Engine (GEE) for near real-time soil moisture estimation in the Florida region.
- Utilized portable microwave radiometer (L-Band) and other sensors to build an R-Shiny application for simulating Brightness Temperature and computing soil moisture in Florida agricultural fields.
- Created a phenology-based algorithm to generate cloud-free and gap-filled images for the entire Florida state.

**Research Scientist** — Sep 2022 – Nov 2024
National Remote Sensing Centre, **Indian Space Research Organization,** Hyderabad

- Performed Land Use Land Cover (LULC) mapping on a 1:250,000 scale using Advanced Wide Field Sensor (AWIFS) sensor data for India.
- Developed a semi-automated model in R and Google Earth Engine (GEE) for LULC mapping.
- Worked on PRISMA hyperspectral data to predict Soil Organic Carbon using Machine Learning and Deep Learning techniques.
- Built a phenology-based thresholding model to map shifting cultivation areas in eastern India using Python and GEE.
- Conducted a workshop at NRSC on Google Earth Engine and its application on resource monitoring for 15 government officials of India.

**Research Fellow** — Feb 2021 – Mar 2022
Centre For Oceans, Rivers, Atmosphere and Land Science, **Indian Institute of Technology Kharagpur**

- Performed field survey and collected information like LAI, DBH, canopy height, and chlorophyll within 200 sample plots.
- Developed a mangrove field inventory for estimation of total carbon stock and chlorophyll.
- Created research methodology and developed a machine learning model in R for mangrove species classification and chlorophyll estimation using imaging spectroscopy.

- Conducted and designed a laboratory course on forest remote sensing for a class size of 18 students.
- Associated with a PhD Scholar in guiding an undergraduate student on mapping mangrove extent over Bhitarkanika mangrove forest, Odisha, India.

**Research Fellow** May 2018 - Sep 2020
Defence Terrain Research Laboratory, **Defence Research and Development Organization**, New Delhi

- Collected more than 5000 spectral data of vegetation, manmade materials, soil, rocks, wood, paints etc. over different regions of India.
- Created a spectral library for classification and feature extraction of material and terrain properties in hyperspectral domain.
- Performed Lab experiments on soil texture and spectral analysis.
- Created a non-parametric model for estimation of soil texture using soil spectral data.
- Monitored flood extent using Google Earth Engine in Northern Bihar, India.
- Tutored 6 trainees on atmospheric correction of Hyperion data.

## INDUSTRY EXPERIENCE

**Freelancer** Mar 2022- Oct 2023

- Developed GEE script for mapping Cameroon's mangroves in 2011 and 2021, and produced a report detailing the methodology, quantitative assessment of restoration areas, and mangrove extent maps.
- Mapped Coastal Wetland Forest (CWF) in Florida and created a habitat suitability map to identify the restoration areas for CWF.
- Identified the vulnerable naalas over the main river stream of the Ganga concerning several natural and anthropogenic phenomena.

**Site Engineer** June 2015 - Nov 2015
Bharat Drilling and Foundation Treatment, Ranchi

- Estimated construction work done and surveyed construction land spread over 80 acres in area.

## RESEARCH INTEREST

- Vegetation dynamics and degradation
- Vegetation biophysical and biochemical parameters
- Soil physics and spectroscopy
- Hyperspectral remote sensing
- Microwave remote sensing
- Reflectance spectroscopy
- Machine learning
- Geospatial Data Science
- Keen to learn and work in LIDAR as well as in UAV data

## TEACHING ASSISTANT

**SWS 3022L: Introduction to Soils in the Environment (Lab)** Fall 2025 and 2026
Soil, Water and Ecosystem Sciences Department, **University of Florida**, USA

- Assisted in teaching undergraduate laboratory sessions (~30 students) on soil physical, chemical, and biological properties.
- Guided students in data collection, analysis, and interpretation using R programming and ArcGIS Pro for soil and environmental applications.

**Forest remote sensing (Lab)** Spring 2021
Centre For Oceans, Rivers, Atmosphere and Land Science, **Indian Institute of Technology Kharagpur**

- Independently designed and delivered a semester-long laboratory course on forest remote sensing for a class of 18 students, developing detailed assignments, grading rubrics, and hands-on exercises
- Guided students through core concepts in vegetation mapping, spectral analysis, and remote sensing data interpretation, adapting instructional strategies to diverse learning needs

**GIS Laboratory** Fall 2017
Department of Remote Sensing, **Birla Institute of Technology, Mesra**, India

- Delivered hands-on GIS laboratory sessions for undergraduate students, guiding them through spatial data acquisition, geodatabase management, and cartographic visualization using ArcGIS and QGIS platforms

## LEADERSHIP

1. Vice President of the Student Committee for the Department of Soil, Water, and Ecosystem Sciences at the University of Florida — Fall 2026-Summer 2027
2. Research Coordinator in Data Science and Informatics club at the University of Florida — Fall 2026-Summer 2017

## PUBLICATIONS

1. **Deep, N.R**., Babaeian, E., Grunwald S., Judge J., Kadyampakeni, D.M., (**2026**), "A Dynamic Effective Emission Depth Framework for Shallow Root-Zone Soil Moisture Retrieval from L-Band Microwave Radiometry in Sandy Soils." Remote Sensing of Environment. (under review).
2. Deep, N.R., Babaeian, E., Grunwald S., (**2026**), "Transferable Physics-Guided Deep Learning Pedotransfer Functions for Soil Hydraulic Property Mapping across Florida." Journal of Hydrology. (under preparation).
3. **Deep, N.R**., Babaeian, E., (**2026**), "Passive Microwave Remote Sensing of Soil Moisture for Agricultural Water Management: Theory, Retrieval, and Applications." In George P. Petropoulos & P. K. Srivastava (Eds.), Microwave Remote Sensing in Environmental Monitoring. Elsevier. **(**In press)
4. **Deep, N.R**., Babaeian, E., Sharma L.K., Grunwald S., Munoz-Carpena R. (**2026**), "Field-Scale Root-Zone Soil Moisture Mapping in Sandy Soils with L-Band Radiometry and Hybrid Radiative Transfer-Machine Learning Modeling." Agricultural Water Management. DOI: https://doi.org/10.1016/j.agwat.2026.110271
5. Babaeian, E., Tuller, M., **Deep, N. R**., Kadyampakeni, D. M., & Jones, S. B. (**2025**), "Field evaluation of the SoilVUE10 time domain reflectometry soil moisture profiling sensor under different installation methods." Vadose Zone Journal. DOI: https://doi.org/10.1002/vzj2.70060
6. Paramanik, S., Behera, M.D., **Deep, N.R.**, Barnwal, S., Bhattacharya, B.K., Behera, S., Dash, J., (**2025**), "Leaf Chlorophyll Concentration Estimation Using Absorption Spectroscopy of AVIRIS-NG for a Mangrove Forest in India." PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science. DOI: 10.1007/s41064-025-00332-3
7. Paramanik, S., **Deep, N.R.**, Behera, M.D., Bhattacharya, B.K., Dash, J., (**2023**), "Species level classification of Mangrove Forest using AVIRIS-NG hyperspectral imagery," International Journal of Remote Sensing and Remote Sensing Letters. DOI: 10.1080/2150704X.2023.2215945
8. Mudi, S., Paramanik, S., Behera, M.D., Prakash, A.J., **Deep, N.R.**, Kale, M.P., Kumar, S., Sharma, N., Pradhan, P., Chavan, M., Roy, P.S., Shrestha, D.G., (**2022**), "Moderate resolution LAI prediction using Sentinel-2 satellite data and indirect field measurements in Sikkim Himalaya." Environmental Monitoring and Assessment. DOI: 10.1007/s10661-022-10530-w

## CONFERENCE PRESENTATIONS

1. **Deep, N.R.**, Babaeian, E., Judge, J., & Grunwald, S., "Estimating Dynamic Effective Emission Depth for Root-Zone Soil Moisture Retrieval Using L-Band Radiometry in Sandy Grassland Soils." CANVAS, Nov 1-4, **2026**.
2. **Deep, N.R.,** Babaeian, E., & Grunwald, S., "High-Resolution Mapping of Soil Water Retention Curves across Florida Using Physics-Guided Deep Learning Pedotransfer Functions." CANVAS Nov 1-4 **2026**.
3. **Deep, N.R.,** Babaeian, E., Grunwald, S., Kadyampakeni, D. M., Sharma, L., & Muñoz-Carpena, R., "A Hybrid UAV L-Band Radiometry and Soil Water Flow Modeling Framework for High Spatiotemporal Root-Zone Soil Moisture Mapping In a Sandy Potato System." CANVAS Nov 1-4, **2026**.
4. **Deep, N.R.,** Babaeian, E., "Spatio-Temporal Downscaling and Forecasting of SMAP Soil Moisture using Hybrid Machine Learning." UF Water Institute Symposium, **2026**.
5. **Deep, N.R.,** Babaeian, E., "Advancing Precision Agriculture: High-Resolution Root-Zone Soil Moisture Retrieval Using UAV-Based L-Band Radiometry." National Soil Moisture Workshop, **2026**.
6. **Deep, N.R.**, Babaeian, E., "Downscaling soil moisture data using google earth engine: demonstrating its efficiency and scalability for high-resolution applications." SWES Research Forum, **2025**.
7. **Deep, N.**, Babaeian, E., Kaur, K., & Sharma, L., "High Resolution Root Zone Soil Moisture Mapping Using a UTV-Mounted L-Band Radiometer over Sandy Soils in Florida." CANVAS, Nov 9-12, **2025**.
8. **Deep, N.**, & Babaeian, E., "Bridging Physics and Machine Learning for Soil Moisture Estimation Using L-Band Radiometry." CANVAS, Nov 9-12, **2025**.

9. Babaeian, E., Rodrigues, H., **Deep, N**., Ghorbani, A., & Grunwald, S. "Physics-Informed CNN-Based Pedotransfer Functions for High-Resolution Mapping of Soil Water Characteristic Curve." CANVAS, Nov 9-12, **2025**.
10. Pujar, G.S., **Deep, N.R**., Chand. K, "Remote sensing based monitoring and assessment of forest vegetation and Plantations during heat waves." International Conference on Heat waves: Forestry and Agriculture under pressure a Remote Sensing Perspective, Oct 19-21, **2023**.
11. **Deep, N.R.**, Pujar, G.S., Singh, R., Murmu, P., Kumar, R., "Comparative analysis of LCLU of Ranchi city and surroundings using multi-sensor remote sensing data with a global land cover product." International Congress of the Indian National Cartographic Association on Emerging Trends in Digital Cartography for Sustainable Ecosystems and Geospatial Economy, Nov 6-8, **2023**.

## AWARD/SCHOLARSHIP

1. Henry Lin Travel Grant, Soil Science Society of America (SSSA) — 2026
2. CALS Doris and Earl Lowe and Verna Lowe Scholarship — 2026
3. Luxmoore Soil Physics Student Travel Award — 2025
4. UF The Water Institute Travel Award — 2025
5. Dr. James Davidson Graduate Travel Scholarship — 2025
6. CALS Graduate Student Council Travel Award — 2025
7. CANVAS Hackathon 2025 – 1$^{St}$ place winner — 2025
8. Secured All India Rank 281 in Graduate Aptitude Test in Engineering (GATE) exam in Geomatics Engineering. — March 2022
9. Minister of Human Resource Development (MHRD) fellowship for PhD — Jan 2022 - Feb 2022
10. Junior Research Fellowship (JRF) from Science and Engineering Research Board (SERB) — Feb 2021 - Jan 2022
11. Junior Research Fellowship (JRF) from Department of Science and Technology (DST) — May 2018 - Sep 2020
12. MHRD Scholarship in M.Tech — 2017

## PROFESSIONAL AFFILIATIONS

1. Life member of the Journal of Indian Society of Remote Sensing (Springer) — Nov 2019
2. Life member of the Journal Indian Society of Geomatics (Springer) — May 2022
3. Member of the Soil Science Society of America (Wiley) — May 2025
4. Member of the American Society for Photogrammetry and Remote Sensing (PE&RS) — Oct 2025
5. Member of the Association of Agricultural Scientists of Indian Origin — Jan 2026

## CERTIFICATIONS/LICENSE

- FAA Drone Pilot License — Aug 2025-Aug 2027

## PROFESSIONAL TRAINING

1. Completed a Massive open online courses (MOOC) on "Spatial Data Science" conducted by ESRI — Oct 2022
2. Participated in MOOC on "Geospatial applications for Disaster risk management" conducted by United Nations Platform for Space based Information for Disaster management and Emergency Response (UN-SPIDER). — Aug 2021
3. Attended several MOOCs conducted by National Aeronautics and Space Administration on water, agriculture and Forestry. — 2019 - 2022
4. Gained certification on "Geospatial data in R" from DataCamp and earned more than 61000 xp. — Dec 2019
5. Participated in International Conference on "Recent Perspective on Climate Change and Sustainable Development" at Suresh Gyan Vihar University, Jaipur, India. — Nov 2019
6. Participated in MOOC on "Introduction to radar remote Sensing" conducted by European Space Agency (ESA). — Oct 2019
7. Participated in ESRI India User Conference 2018 at New Delhi, India. — Sep 2018
8. Attended short-term training on "Hyperspectral Science and Applications" at Space Application Centre, Indian Space Research Organization (ISRO), India. — Aug 2018
9. Participated in ISRO outreach program on "Basics of Remote Sensing, Geographical Information System & Global Navigation Satellite System". — Nov 2016

## SKILLS RELEVANT TO RESEARCH

**Programming skills**

- Google Earth Engine (JavaScript) and R for data analysis, visualization and modelling
- Python expertise as an intermediate level

**Data Science**

- Machine Learning, Deep Learning, SQL and Excel

**GIS Software**

- ArcGIS 10.X desktop, ArcGIS Pro, ENVI, ERDAS (licensed)
- QGIS, SNAP, Global Mapper, Google Earth Pro (open source)

**Field Instruments**

- LAI-2200C Plant Canopy Analyzer
- Laser Range Finder
- Spectroradiometer (350 – 2500 nm)
- Portable Microwave Radiometers (L-band)
- SPAD
- L-Band radiometer
- PR2 sensor
- Acclima TDR
- Apogee NDVI
- Apogee TIR
- Pika-L Resonon Hyperspectral Camera

## EXTRA-CURRICULAR/VOLUNTEER/HOBBIES ACTIVITIES/SERVICES

1. Singing and travelling
2. Cricket and Adventure sports

## LANGUAGES

1. English – Proficient in reading, listening, writing and speaking (**IELTS – 7**)
2. Hindi – Native language

## REFEREES

1. Dr. Ebrahim Babaeian
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   University of Florida, USA
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